

PYTHON FOR EVERYBODY

A Comprehensive Learning Companion

Course 1

Getting Started with Python

Variables • Expressions • Conditionals • Functions • Loops

Intermediate Level | Python 3 | Comprehensive Theory & Hands-on Coding

Programming as Language Acquisition

Vocabulary
(Words)

Variables &
Reserved Words

Grammar
(Syntax)

Strict combinatory
rules (e.g., $x = x + 2$)

Story
(Program)

Meaningful sequence
to achieve a goal

Python prioritises **human readability**,
minimising syntactic overhead to drastically
lower the cognitive barrier to entry.

visual_code_editor.py

```
1  # The Vocabulary
2  greeting = 'Hello, World'
3
4  # The Grammar & Story
5  if greeting == 'Hello, World':
5  |     print(greeting)
```

Defining Variables
& Concepts

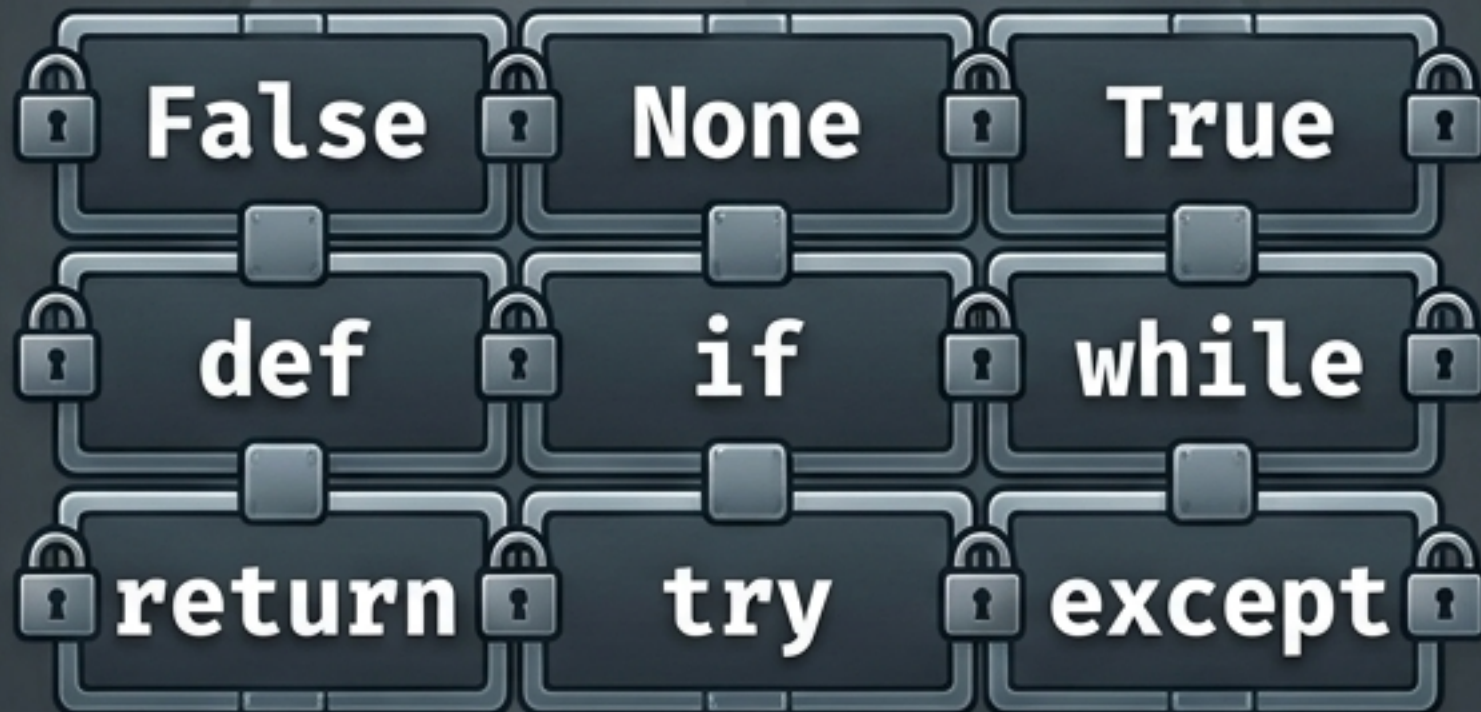
Implementing
Logic &
Execution Flow

Conditional Logic
(Grammar)

Function Call
(Action)

The Forbidden Dictionary & Variable Naming

The 35 Reserved Words



Exclusively owned by the Python interpreter.
Cannot be used as names.

Custom Variable Rules

- ✓ **Must begin with a letter or underscore (`_`), never a number.**
- ✓ **Combinations of letters, digits, and underscores permitted thereafter.**
- ✓ **Strictly case-sensitive:** 'spam', 'Spam', and 'SPAM' are distinct.
- ✓ **Best Practice:** Use mnemonic, descriptive names (e.g., 'hours_worked').

The Four Fundamental Building Blocks

1



Variables: Named storage containers whose values change dynamically.

2



Operators: Symbols that perform calculations or logic.

3



Constants: Fixed, literal values written directly into code.

4



Functions: Named, reusable blocks of prepackaged code.

Variable

Operators

```
age = 25 + calculate_years()
```

Constant

Function

Python's Core Data Types

Periodic Table of Data Matrix		
Type	Description	Examples
int	Whole numbers of arbitrary magnitude without a decimal point.	0, -42, 1000000
float	Numbers with a fractional component stored in double-precision format.	3.14, 2.0, -0.001
str	Immutable sequences of characters delimited by single or double quotes.	'Python', "hello"
bool	Exactly two instances representing truth logic. A subclass of int.	True (1), False (0)
NoneType	The singleton constant representing the intentional absence of a value.	None

Numeric Operators & The Division Quirk

Addition: +

Multiplication: *

Modulo (Remainder): %

Subtraction: -

Exponentiation: **

Historical Division: Python 2 vs Python 3

/

7 / 2 → 3.5

Always returns a floating-point result
in Python 3.

//

7 // 2 → 3

Discards the fractional part, returning an
integer. Resolves silent data loss issues from
older systems.

Operator Precedence (BODMAS / PEMDAS)

1. Parentheses: ()

2. Exponents: **

3. Multiplication, Division,
Modulo: *, /, //, %

4. Addition, Subtraction: +, -

The Power of Parentheses

Parentheses override default precedence.

$$2 + 3 * 4 = 14$$

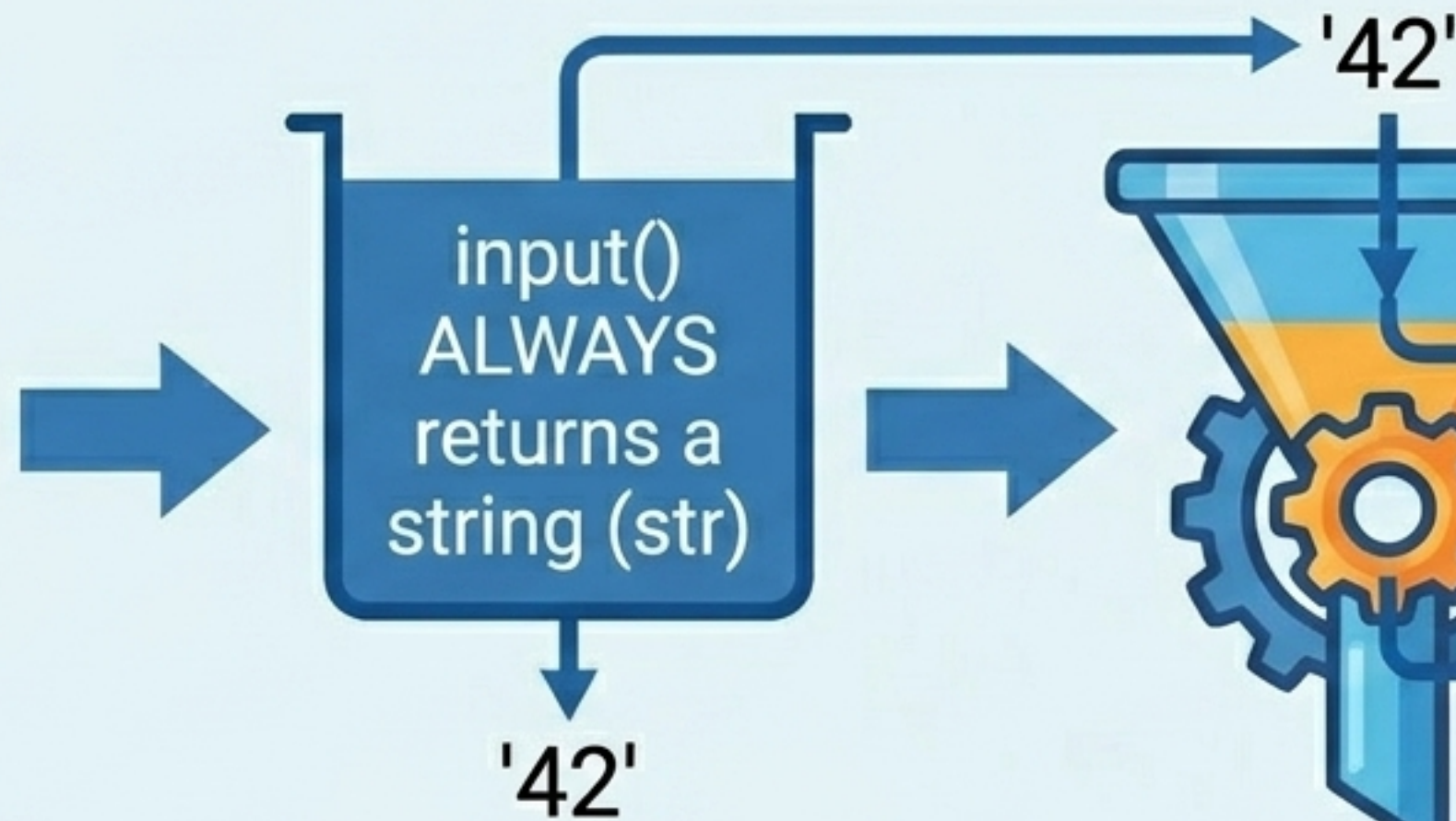
$$(2 + 3) * 4 = 20$$

User Input & Type Conversion

External World



The input() Function



Type Conversion Filter

Explicit type casting is required to perform maths.

Path A

`int('42') → 42`
(Whole number)

Path B

`float('42') → 42.0`
(Fractional number)

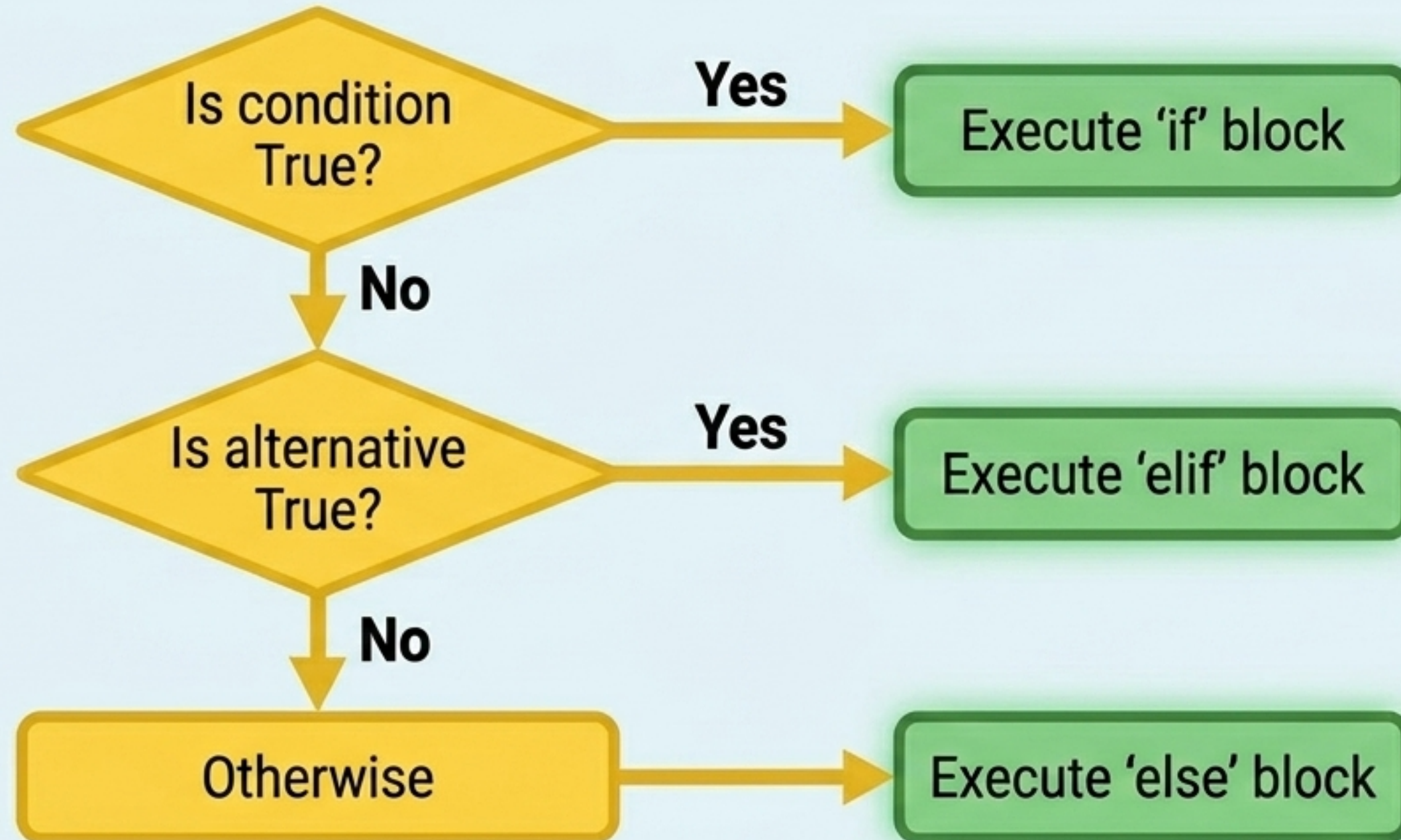
Indentation: Visualising Logic

- Unlike other languages that use brackets {}, Python uses visual indentation to group statements.
- Indentation is grammatically mandatory; incorrect spacing triggers an `IndentationError`.
- A colon (:) signals that an indented code block is about to begin.
- Statements aligned at the exact same indentation level form a single cohesive block.

```
temperature = 22

if temperature > 20:
    print('It is warm.')
    print('Wear short sleeves.')
print('End of forecast.')
```


Conditional Statements (Decision Making)

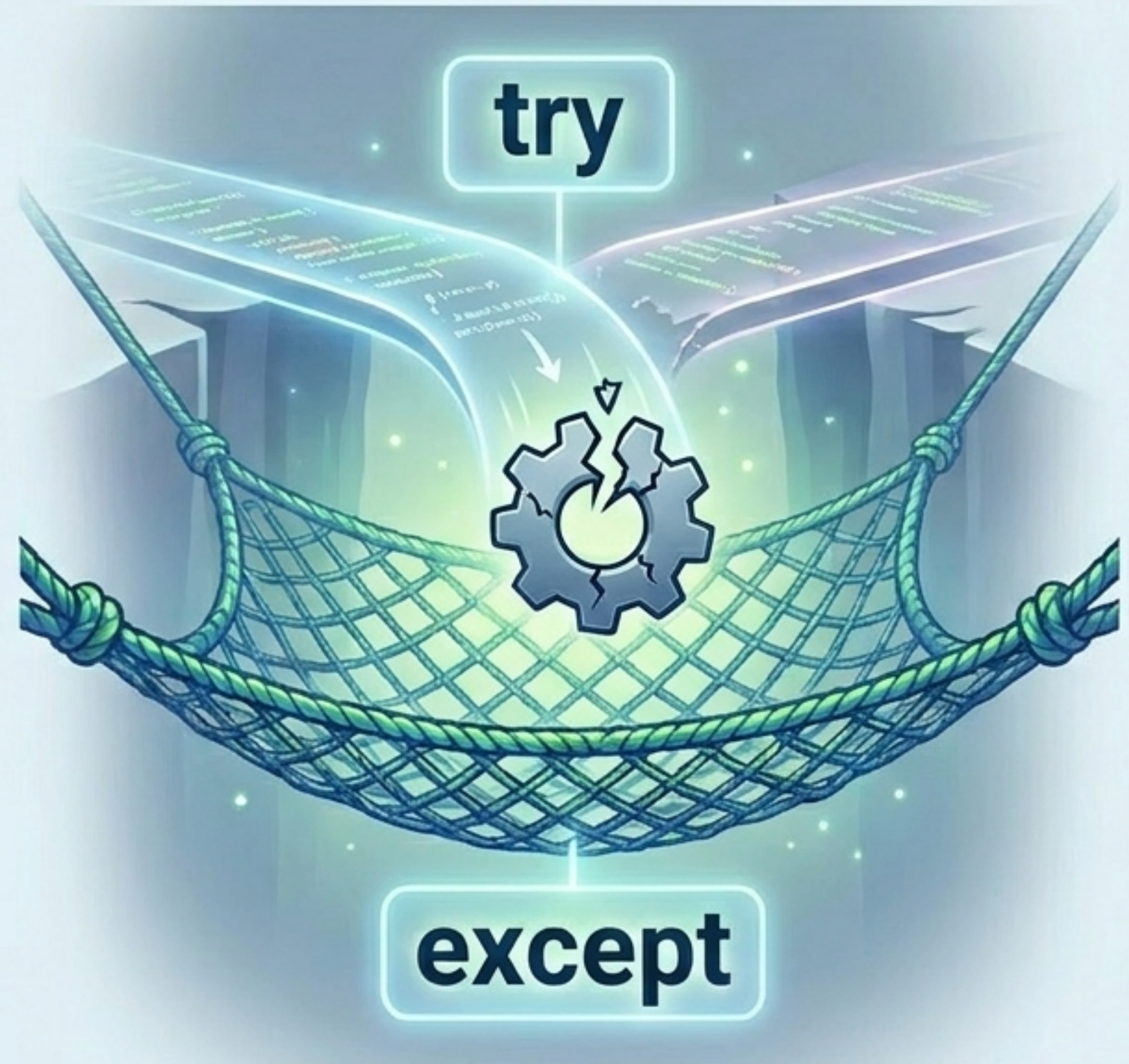


💡 Only one block in an if-elif-else chain will ever execute per pass.

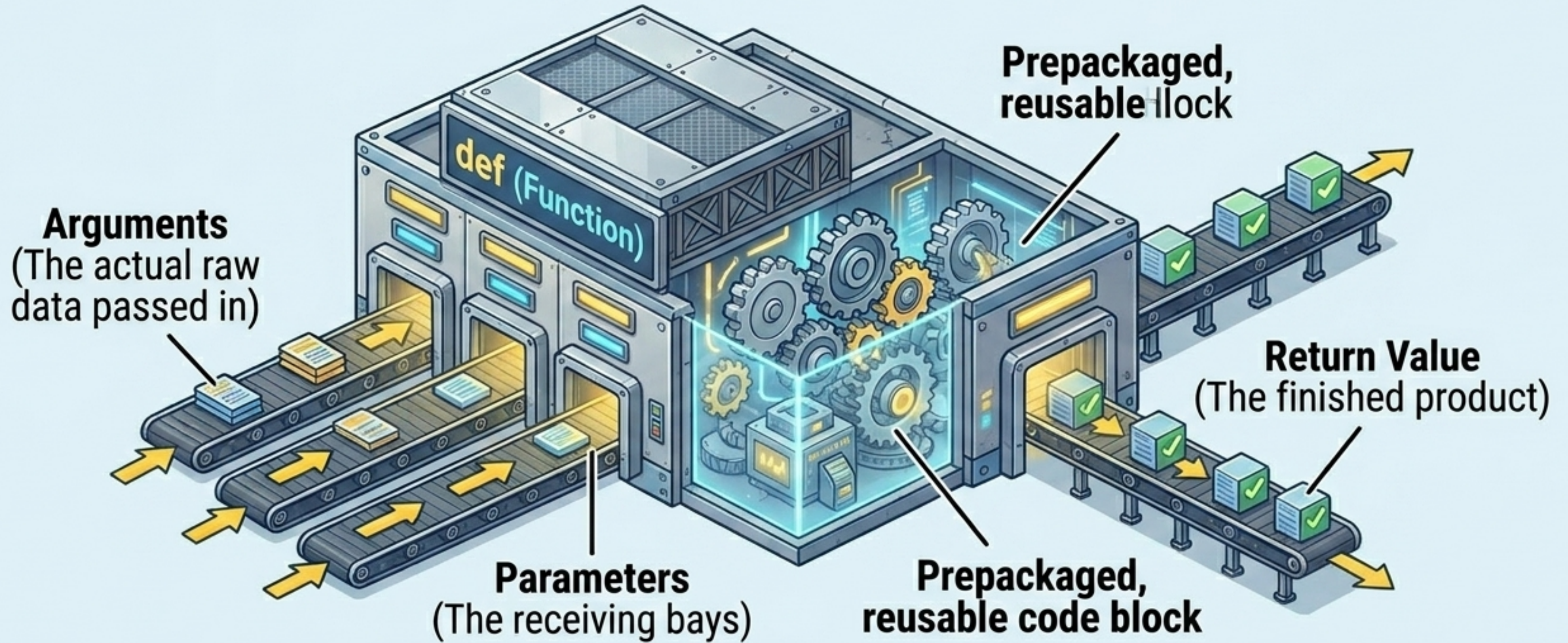
Handling Errors Gracefully

The try-except Construct

- When Python encounters an invalid operation (e.g., dividing by zero), it halts execution and crashes.
- The 'try' block lets you test a risky section of code.
- If an error occurs, Python immediately jumps to the 'except' block instead of crashing.
- The 'except' block acts as a safety net, allowing the program to recover, log the issue, or prompt the user again.



Functions: The Mini-Factories



Defining a function registers it in memory; it does not execute the code until it is explicitly 'called' by its name.

Function Return Values

The 'return' Statement & Computed Data

- The 'return' statement immediately ends execution and hands a computed value back to the caller.
- Unlike `print()`, which simply displays text to a human screen, `return` passes actual data that can be stored in a variable.
- If a function completes without a `return` statement, it implicitly returns 'None'.
- Variables created inside a function are temporary (local scope).



The wrong way (Showing to human)

```
def add(a, b):  
    print(a + b)
```

```
result = add(2, 3)  
print(result) # Output:  
None
```



The right way (Handing back to machine)

```
def add(a, b):  
    return a + b
```

```
result = add(2, 3)  
print(result) # Output:  
5
```



Loops and Iteration: Automating Repetition

 while Loops (Indefinite)	 for Loops (Definite)
Executes continuously as long as a specified condition remains True.	Iterates over a known, predefined sequence of items (like a string or list).
Waits for the environment or a variable state to change.	Automatically handles progression through the sequence.
Risk: Requires careful management to avoid infinite loops that crash the system.	Stops precisely and safely when the sequence reaches its end.

Course Summary: 9 Fundamental Rules

1



Python mirrors **natural language**: Variables are vocabulary, statements are **sentences**.

2



Variable names are **case-sensitive** and cannot use the 35 reserved keywords.

3



Python assigns **variables dynamically**; their types can change during execution.

4



Explicit **type conversion** (``int``, ``float``, ``str``) is required for mismatched data.

5



The ``input()`` function **always** returns a **string**, regardless of what the user types.

6



``//`` performs floor division; ``/`` guarantees a floating-point true division result.

7



Indentation is grammatically mandatory and defines logical code blocks.

8



Functions (``def``) store reusable logic; ``return`` hands the computed data back.

9



``for`` loops iterate a specific number of times; ``while`` loops run until a condition turns False.